Solution

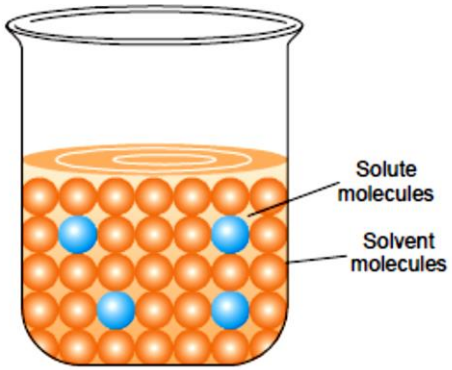
Solution is a homogeneous mixture of two or more substances on molecular level.

Or

A solution is a homogeneous mixture of Solute and solvent.

Or

A solution is a homogeneous mixture of two substances but consisting of one phase.

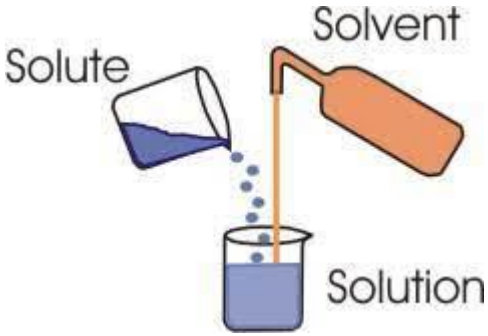


Molecular model of a solution

The constituent of the mixture present in a smaller amount is called the **Solute** and the one present in a larger amount is called the **Solvent.**

The Substance which is dissolved in a liquid is called "Solute".

The Liquid in which Solute is dissolved is known as "Solvent".



For example, when a smaller amount of **sugar (solute)** is mixed with **water (solvent),** a homogeneous solution in water is obtained. In this solution, sugar molecules are uniformly dispersed in molecules of water. Similarly, a solution of salt (Na^+, Cl^-) in water consists of ions of salt (Na^+, Cl^-) dispersed in water.

Properties of Solution:

- A solution is a homogeneous mixture.
- The size of a solute particle in a solution extremely small. It is less than 1 nm in diameter (1 nanometer=10^{-9}metre).
- The particles of a solution cannot be seen even with a microscope.
- The particles of a solution pass through the filter paper. So, a solution cannot separated by a filtration.
- The solutions are very stable. The particles of solute present in a solution do not separate out on keeping.
- A true solution does not scatter light.
- It is composed of only one phase.

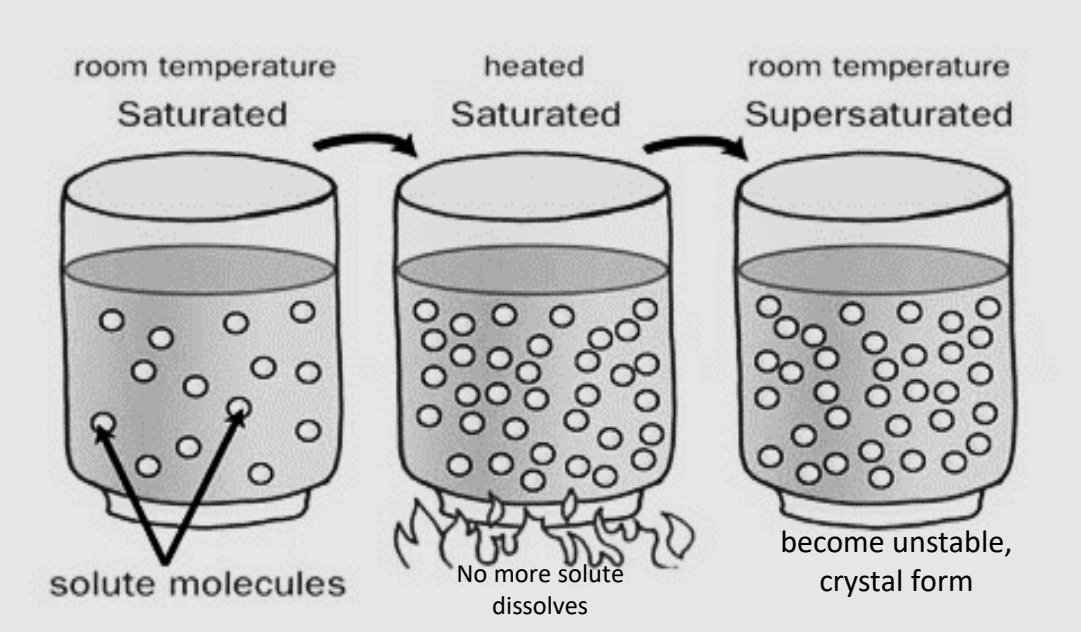
Types of Solutions

The common solutions that we come across are those where the solute is a solid and the solvent is a liquid. In fact, substance in any three states of matter (solid, liquid, gas) can act as solute or solvent. Thus there are **seven types of solutions whose examples are listed in Table:**

Types and Examples of Solutions		
State of Solute	**State of Solvent**	**Example**
Gas	**Gas**	**Air**
Gas	**Liquid** (Carbonated drinks)	**Oxygen in water, CO_2 in water**
Gas	**Solid**	**Adsorption of H_2 by palladium**
Liquid	**Liquid**	**Alcohol in water**
Liquid	**Solid**	**Mercury in silver**
Solid	**Liquid**	**Sugar, Salt**
Solid	**Solid** (Steel)	**Metal alloys : Carbon in iron**

Types of Solutions

1. Unsaturated: An unsaturated solution is a solution in which more solute can be dissolved at a given temperature.
2. Saturated: A saturated solution is a solution containing the maximum amount of solute that will dissolve at a given temperature.
3. Supersaturated: A supersaturated solution is a solution that contains more solute than would dissolve in a saturated solution at a given temperature.



Concentration of a Solution

The concentration of a solution is defined as : **the amount (mass or volume of solute present in a given amount of solvent or solution**.

For example: A 10% aqueous solution of common salt contain 10g NaCl per 100g of solution. i.e. 10 g NaCl dissolved in 90 g water.

Concentration is generally expressed as the quantity of solute in a unit volume of solution.

$$\text{Concentration} = \frac{\text{Quantity of solute}}{\text{Volume of solution}}$$

A solution containing a relatively low concentration of solute is called **Dilute solution.**

A solution of high concentration is called **Concentrated solution.**

Ways of Expressing Concentration

The amount of solute, solvent and solution may be measured by volume, weight or number of moles. Therefore, concentration of a solution can be measured in many ways.

(a) Percentage

(b) Mole fraction

(c) Molarity

(d) Molality

(e) Normality

Percentage (%)

This is based on mass (m) or volume (v) of components of solution. It is of four types.

i. Percentage in $\frac{M}{M}$ $\left[\frac{mass}{mass}\%\right]$

ii. Percentage in $\frac{M}{V}$ $\left[\frac{mass}{vol}\%\right]$

iii. Percentage in $\frac{V}{M}$ $\left[\frac{vol}{mass}\%\right]$

iv. Percentage in $\frac{V}{V}$ $\left[\frac{vol}{vol}\%\right]$

For example,

i. 5% (M/M) solution means solute 5 g in 95 g solvent

ii. 10% (M/V) solution means solute 10 g in 100 cm^3 solution.

iii. 5% (V/M) solution means solute 5 cm^3 in 100 g solution.

iv. 15% (V/V) solution means solute 15 cm^3 in 85 cm^3 solvent.

Example: What is the percent by weight of NaCl if 1.75 g of NaCl is dissolved in 5.85 g of water?

Solution:

Wt. of solute (NaCl) = 1.75 g

Wt. of solvent (H_2O) = 5.85 g

∴ Wt. of solution = 1.75 + 5.85 = 7.60 g

Hence concentration of NaCl % by weight

$$= \frac{1.75}{7.60} \times 100$$

$$= 23.0$$

Example: 10g of NaCl is dissolved in 50 g of water. Calculate percentage (%) of solution?

MIXTURES

• All solutions are mixtures.

•Mixtures are of two types ;

A) Homogeneous Mixture

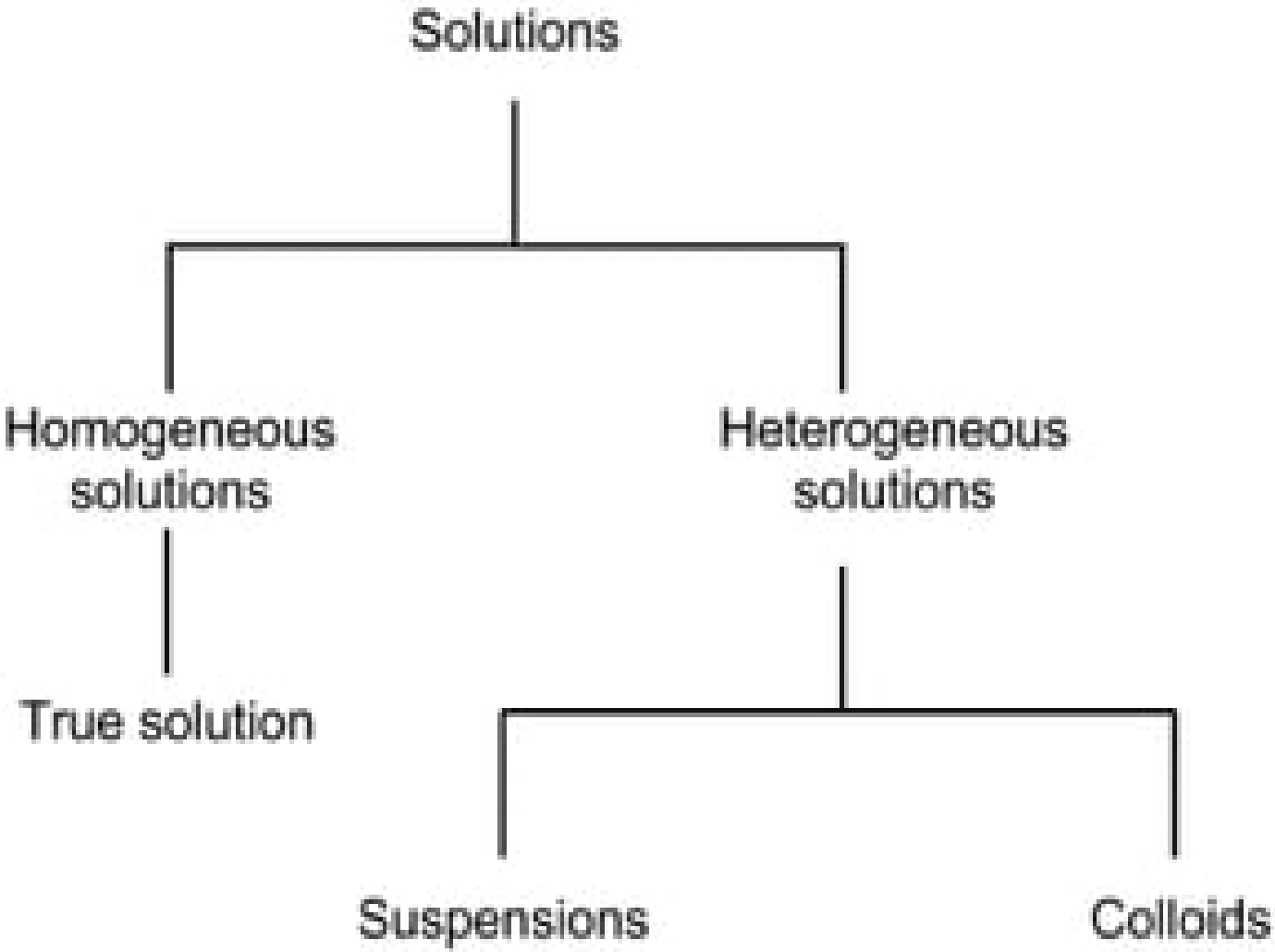
A mixture which has uniform composition and properties throughout. For example, air is a homogeneous mixture of gases. A teaspoonful of table salt stirred into a glass of water also makes a homogeneous mixture.

B) Heterogeneous Mixture

A heterogeneous mixture is simply any mixture that is not uniform in composition – it's a non-uniform mixture of smaller constituent parts. Using various means, the parts in the mixture can be separated from one another.

Example: Cereal in milk is an example of a heterogeneous mixture. Soil is another example. Soil has pebbles, plant matter and

sand in it.


Solutions
Homogeneous solutions
Heterogeneous solutions
True solution
Suspensions
Colloids

TRUE SOLUTION

A true solution is a homogeneous solution in which the solute particles have diameters less than 10-7cm. i.e., the solute particles are of molecular dimensions. The particles are invisible even under powerful microscopes. For example, sodium chloride in water is a true solution.

COLLOID

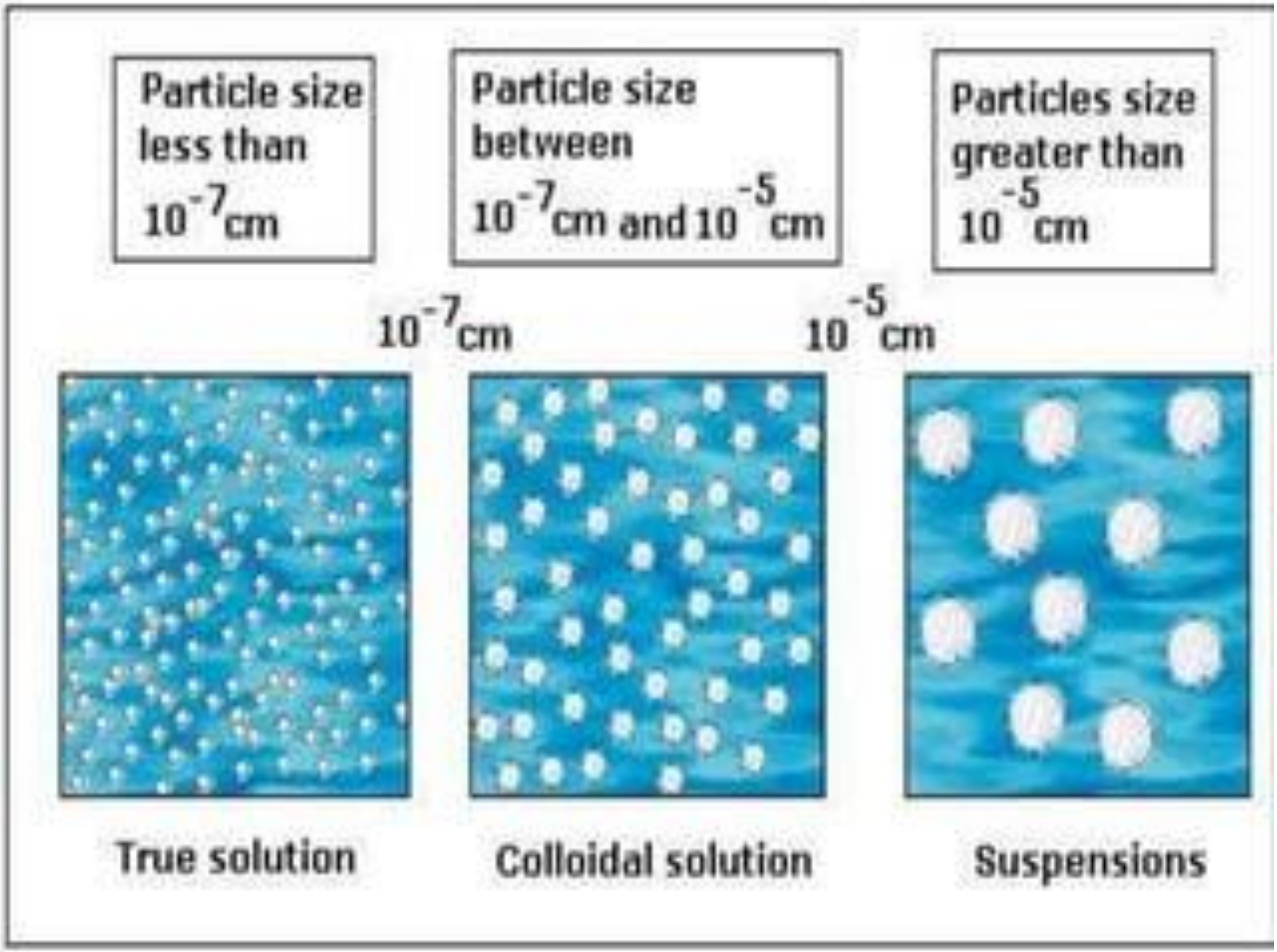
A colloid is a solution that has particles ranging between 1 and 1000 nanometers in diameter, yet are still able to remain evenly distributed throughout the solution. These are also known as colloidal dispersions because the substances remain dispersed and do not settle to the bottom of the container.

SUSPENSION

A suspension is a heterogeneous mixture in which solute-like particles settle out of a solvent-like phase some time after their introduction.

We apply the word 'suspension' when particles are big enough to eventually settle. We use the terms 'solute-like' and 'solvent- like,' because we are dealing with a heterogeneous mixture, while the terms solute and solvent refer to solutions, which

are homogeneous.


Particle size less than 10^{-7}cm
Particle size between 10^{-7}cm and 10^{-5}cm
Particles size greater than 10^{-5}cm
10^{-7}cm
10^{-5}cm
True solution
Colloidal solution
Suspensions

COLLOIDS

• A colloid is a kind of solution in which the solute particles is intermediate between those a true solution and those in suspension.

• Examples: soap solution, starch solution, milk ink, blood, jelly and solutions of synthetic detergents.

• Colloids are also know as Colloidal solution.

PROPERTIES OF COLLOIDS

• A colloid is a heterogeneous mixture.

• The size of particles of a colloid is too small to be individually seen by naked eyes.

• Collides are big enough to scatter a beam of light passing through it and make its path visible.

• They do not settle down when left undisturbed , that is, a colloid is quite stable.

SUSPENSION

• A Suspension is a heterogeneous mixture in which the small particles of a solid are spread throughout the liquid without dissolving in it.

• Examples: chalk-water, muddy water, milk of magnesia, sand particles suspended in water, flour in water.

• Those substances which are insoluble in water forms suspension.

PROPERTIES OF SUSPENSION

• A suspension is a heterogeneous mixture.

• The size of a solute particle in a suspension are quite large.

• The particles of a suspension can be seen easily.

• The particles of a suspension do not pass through the filter paper. So, a suspension can separated by filtration.

• The suspension is unstable. The particles of solute present in a suspension settle down after some time.

• A suspension scatters a beam of light passing through it.

Moles

When the amount of solute is given in grams and its molecular weight is MW, it can be converted to moles :

mole = mass / molecular weight

$$n = x\,g \times \frac{1\,mol}{MW\,g}\,mol$$

Example: How many grams of N_2 are in 0.400 moles N_2

Mole Fraction

A simple solution is made of two substances : one is the solute and the other solvent. Mole fraction, *X, of solute is defined as* ***the ratio of the number of moles of solute and the total number of*** **moles of solute and solvent. Thus,**

$$X_{solute} = \frac{Moles\ of\ solute}{Moles\ of\ solute + Moles\ of\ solvent}$$

If n represents moles of solute and N number of moles of solvent,

$$X_{solute} = \frac{n}{n + N}$$

Notice that mole fraction of solvent would be

$$X_{solvent} = \frac{N}{n + N}$$

Mole fraction is unitless and *Xsolute + Xsolvent = 1*

Molarity

In current practice, concentration is most often expressed as molarity.

Molarity (symbol *M) is* defined as

the number of moles of solute per litre of solution.

If n is the number of moles of solute and V litres the volume of solution,

$$Molarity = \frac{Moles\ of\ solute}{Volume\ in\ litres}$$

$$M = \frac{n}{V\ (in\ liters)}$$

For one mole of solute dissolved in one litre of solution *M = 1 i.e., molarity is one. Such a* solution is called 1*M* .

A solution containing two moles of solute in one litre is 2M.

Unit of molarity is mol litre^{-1}.

Molality

Molality of a solution (symbol *m) is defined as* ***the number of moles of solute per kilogram of*** **solvent**

$$\text{Molality (m)} = \frac{\text{Moles of solute}}{\text{Mass of solvent in kilograms}}$$

A solution obtained by dissolving one mole of the solute in 1000 g of solvent is called **one molal** or **1*m solution.***

Difference between molality and molarity:

Molality is defined in terms of mass of solvent.

Molarity is defined in terms of volume of solution.

Normality

Normality of a solution (symbol N) is defined as number of equivalents of solute per litre of the solution .

$$\text{Normality (N)} = \frac{\text{Equivalents of solute}}{\text{Volume of solution in Litres}}$$

Thus, if **40 g of NaOH** (eq. wt. = 40) be dissolved in **one litre** of solution, normality of the solution is one and the solution is called **1N (one-normal)**.

A solution containing 4.0 g of NaOH is 1/10 N or 0.1 N or decinormal.

Math 1: How many grams of N_2 are in 0.400 moles N_2

Math 2: What weight of HCl is present in 155 ml of a 0.540 M solution ?

Math 3: What is the molarity of a solution prepared by dissolving 75.5 g of pure KOH in 540 ml of solution?

Math 4: What is the molality of a solution prepared by dissolving 5.0 g of toluene (C_7H_8) in 225 g of benzene (C_6H_6)?

Math 5: 5 g of NaCl is dissolved in 1000 g of water. If the density of the resulting solution is 0.997 g per ml, calculate the molality, molarity, normality and mole fraction of the solute, assuming volume of the solution is equal to that of solvent.

Henry's Law

In a particular solvent, the relationship between the pressure and solubility of gas was given by William Henry in 1803. This generalization is known as Henry's law, which states that when a gas is dissolved in a solvent at a constant temperature, the concentration of the gas is directly proportional to the pressure of the gas.

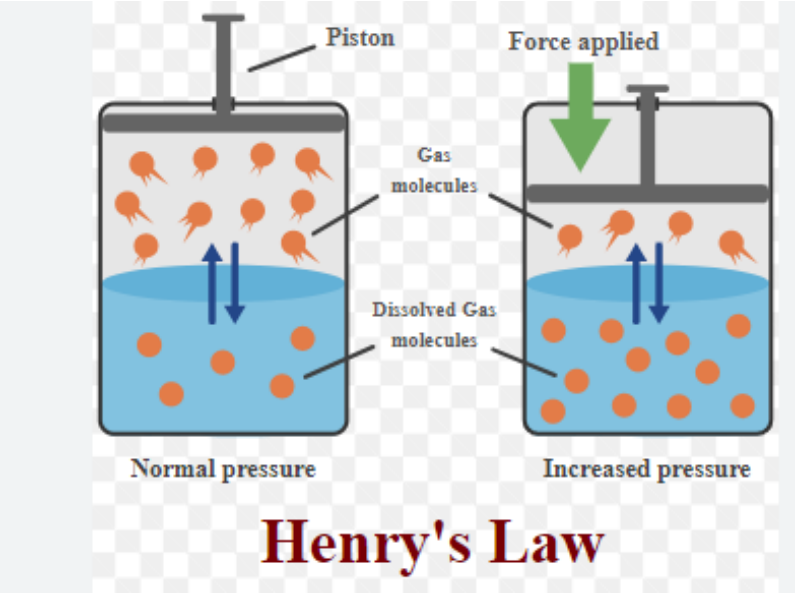
The gas solubility in a solvent depends on the pressure and temperature. When a gas is enclosed in a container over its saturated solution, an equilibrium is established between the molecules of gas and the gas in the solution.

$$\text{Gas} \rightleftharpoons \text{Gas in solution}$$

When the pressure is increased in the system, according to Le Chatelier Principle, this results in shifting the equilibrium to the direction where pressure is decreased. The pressure can be further decreased by dissolving more gas in the solvent. However, the concentration of a gas in a solvent can be increased by increasing the pressure.

Henry's Law :

At constant temperature, the amount of a given gas dissolved in a given type and volume of liquid is directly proportional to the partial pressure of that gas in equilibrium with that liquid .



The mathematical formula of Henry's law is given by:

$$P \alpha C \text{ or } P = k_H \times C$$

When liquid and gas are combined in a closed vessel,

a dynamic equilibrium exit between the gas molecules

And those in liquid.

Gas $\Longleftrightarrow$ Gas in liquid

As the pressure of the gas above the liquid increased double,

the gas molecules become more soluble in the solvent.

Henry's law constant

Mathematically, Henry's law may be expressed as,

$C \propto P$

or

$C = k \times P$

$$P = k_H \times C$$

$$C \propto 1/k_H$$

When, $P = 1$

Where, C = concentration of the gas in a solution

k = proportionality constant

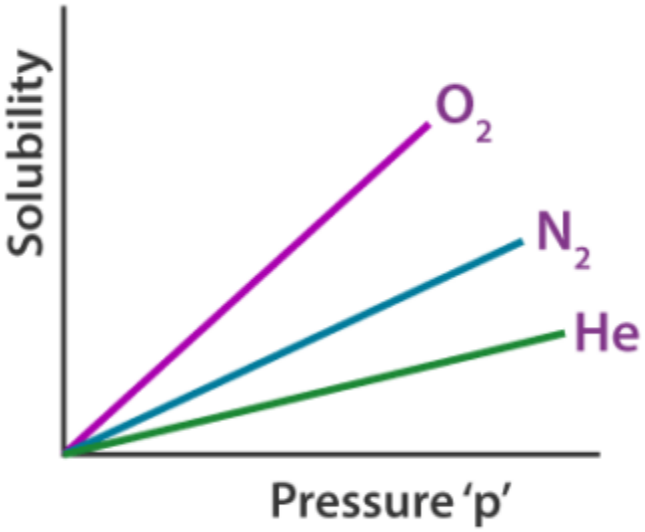
P = pressure of the gas

The above expression is also known as Henry's law equation.

Thus, the K_H is inversely proportional to the solubility(C).So, a small value of K_H indicates the high solubility whereas ,a high value indicates the low solubility.

Factors affecting Henry's law constant

Henry law constant is represented by "k" and its value depends on the following factors which are:



- Nature of the gas
- Nature of the solvent
- Units of pressure
- Units of concentration
- Temperature and pressure of the system

The nature of gas and solvent has a great effect on Henry's law constant. This is a reason, different solvents and gases have different values of Henry's law constant

Applications of Henry's law

These are the following applications and uses of Henry law:

- In the production of carbonated drinks
- In the service of cask-conditioned beer
- In respiration and blood oxygenation
- In underwater diving

Production of carbonated drinks

Hendry's law is used to dissolve pure carbon dioxide (CO_2) in carbonated drinks. The pressure of carbon dioxide (CO_2) is usually more than the standard pressure in the bottles of carbonated drinks.

However, when the bottles are opened, carbon dioxide comes out with pressure and makes a hissing sound. This results in an instant decrease in the partial pressure of carbon dioxide (CO_2) in the atmosphere in the drink's bottle. According to henry's law, the solubility of CO_2 also decreased. This is the reason carbon dioxide (CO_2) moves to the surface of the drink in the bottle and forms bubbles and is released into the atmosphere.

When the carbonated drink is left open for several minutes, an equilibrium is established between the carbon dioxide (CO_2) in the atmosphere and the carbon dioxide (CO_2) in the drink. Due to this, carbonated drinks lose their fizzy taste.

Respiration and blood oxygenation

This principle plays an important role in respiration. When air is inhaled, it increased the partial pressure of oxygen (O_2) in the alveoli of the lungs. This is due to the interaction between deoxygenated blood and oxygen-rich air molecules. Similarly, during exhalation, the partial pressure of carbon dioxide (CO_2) is increased.

These are the following gas exchange reactions:

- Deoxygenated blood has a lower amount of dissolved oxygen while alveoli have a larger amount of oxygen that is came from the air. Therefore, the partial pressure of oxygen in the alveoli is high. However, oxygen moves from the alveoli to the deoxygenated blood.
- Similarly, the amount of carbon dioxide (CO_2) is very high in the deoxygenated blood and lower in the alveoli. This results in an increase in the partial pressure of carbon dioxide in the deoxygenated blood. So, carbon dioxide moves from the deoxygenated blood to the alveoli of the lungs.

Underwater diving

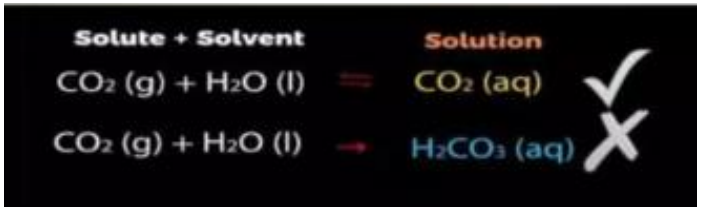
According to henry’s law, increased pressure means solubility of more gases. Therefore, at the depth of the sea, divers consume more gases because the pressure is more at the bottom of the sea as compared to the surface of the sea.

When divers breathed compressed air into their lungs, both the oxygen and nitrogen dissolve in the bloodstream. When they ascend to the surface where pressure is normal, nitrogen gas bubbles out of the solution and causes painful symptoms of decompression sickness in the body of divers.

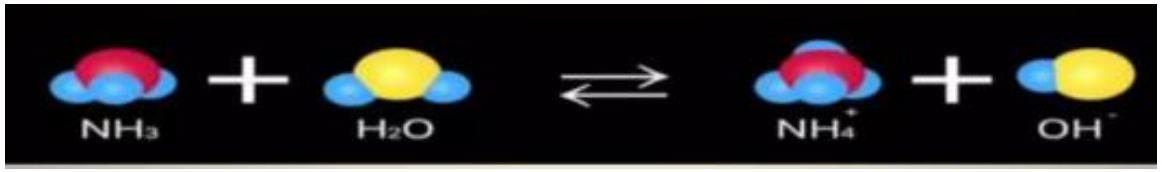
Limitations

Henry's law applies only to those gases that have nearly ideal behaviors. These are the limitations of Henry's law, it is applicable when:

- The temperature and pressure of the system are moderate.
- The solubility of the gas in a solvent is low.
- Gas does not react with the solvent to form a new species.



- Association and dissociation of the gas do not take place.



- The system is in equilibrium.

Problem:

1. Calculate the solubility of gaseous oxygen in water at a temperature of 293 K when the partial pressure exerted by O_2 is 1 bar. (Given: k_H for O_2 34840 bar.L.mol^{-1})

2. The value of k_H for carbon dioxide at a temperature of 293 K is $1.6*10^3$ atm.L.mol^{-1}. At what partial pressure would the gas have a solubility (in water) of $2*10^{-5}$ M?